

Question ONE

- a) Explain the following terms as applied to instrumentation
 i) Range ii) span iii) Repeatability iv) Reproducibility (4 marks)
- b) i) Give any TWO reasons why platinum material is preferred to copper in resistance temperature detectors(RTDs)
- ii) A platinum resistance thermometer has a resistance of 100Ω at 20°C . Determine the resistance at 55°C given the temperature coefficient of resistance of platinum (α) at 20°C is $0.00394/^\circ\text{C}$. Assume a linear resistance-temperature relationship.
- iii) The platinum resistance thermometer in part ii was incorporated in a bridge whose resistance in the three arms is 100Ω . The bridge is supplied by a 12 V supply, determine the offset voltage (6marks)
- i) Explain why the material used in photoconductive transducers is made thin and wound back and forth

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Page 1 of 6

- ii) A capacitor transducer is made up of two concentric cylindrical electrodes. The outer diameter of the inner electrode is 6mm and the inner diameter of the outer electrode is 36mm . Calculate the change in capacitance if the inner electrode is moved a distance of 5mm . (Take permittivity of free space is $8.85 \times 10^{-12}\text{ F/m}$)
- d) Explain the operation of the arrangement shown in Figure Q1d (5 marks)

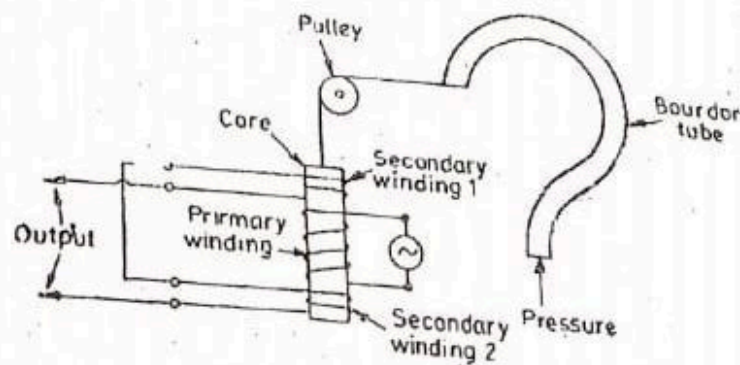


Figure Q1d

Question TWO

- c) In a measurement system to measure pressure, the diaphragm is deflected through $0.5 \times 10^{-3} \text{ mm}$ when a pressure of 1000 N/m^2 is applied. The diaphragm is connected to an LVDT which has an output of 6 V rms when the displacement is $0.4 \times 10^{-3} \text{ mm}$. The output of the LVDT is connected to a 10 V voltmeter with 100 scale divisions $\frac{1}{100}$ of division can be estimated with care. Determine

- The sensitivity of the diaphragm
- the sensitivity of the LVDT
- the overall sensitivity of the measurement system
- the resolution of the whole system

(6 marks)

- d) A piezo electric crystal having a thickness of 8 mm and a voltage sensitivity of $0.055 \text{ V m}^2/\text{N}$. The material has absolute permittivity of $56 \times 10^{-12} \text{ F/m}$. The output voltage is 150 V , determine

- the pressure applied
- Charge sensitivity

(3 marks)

Question THREE

- a) Explain the following terms

i) piezo-resistive ii) piezoelectric iii) Seebeck effect (3 marks)

- b) Derive the expression relating the output voltage V_o to the displacement x of the material of applied force on piezoelectric transducer having a thickness $t \text{ mm}$ (4 marks)

- c) ✓ The following data was obtained from the tensile test on a strain gauged rod;

- Original resistance of gauge $= 450 \Omega$
- Final resistance of the gauge $= 450.35 \Omega$
- Gauge factor $= 2$
- Modulus of Elasticity of the rod $= 200 \text{ GN/m}^2$
- Diameter of rod $= 14 \text{ mm}$

Determine the tensile load applied on the strain gauge (5 marks)

- d) i) State the Law of intermediate temperatures as applied to thermocouples
ii) A type J thermocouple was used to measure the temperature of a liquid (T_1). The temperature of the reference junction was maintained at 30°C . The emf was found to be 7.545 mV . Use the thermocouples tables provided to find the temperature of the liquid. (5 marks)

- e) ✓ Distinguish between the following transducer types

- passive and active
- primary and secondary

Page 3 of 4

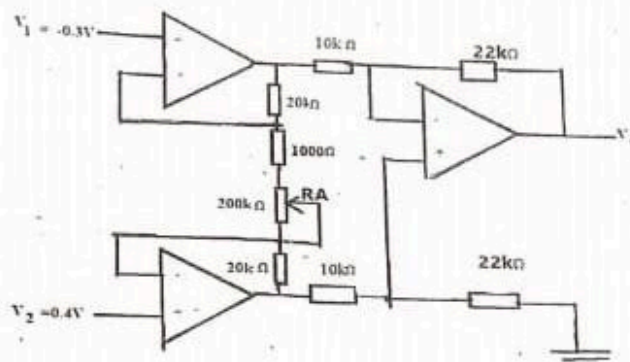


Figure Q4c

d) Draw the block diagram to show the elements of a instrumentation system and explain the function of each component

Question FIVE

- Explain any THREE advantages of electronic instrumentation system over the mechanical types (3 marks)
- Explain any TWO applications of instrumentation systems (2 marks)
- An instrument system to convert temperature into a Digital output is shown in Figure Q5c. The transducer has a sensitivity of $7.5\text{mV}/^\circ\text{C}$ and must measure up to 100°C . The 6 bit ADC is used with 12 volt reference. Determine
 - the ADC input voltage corresponding to full scale output ✓ $V_{in} = V_{ref}$
 - the gain of the amplifier 2^n
 - analogue voltage corresponding to 1 bit change in the ADC (6 marks)

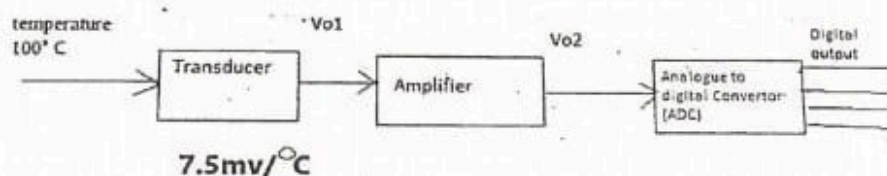


Figure Q5c

- Draw a diagram of an x-y recorder and explain its operation (9 marks)

Question FOUR

- a) For the equal arm bridge circuit of Figure Q4a, a thermistor is connected to one of the arms. Derive the expression for the output voltage when the temperature is applied, the thermistor resistance increases to $R+\Delta R$ (4 marks)
- b)

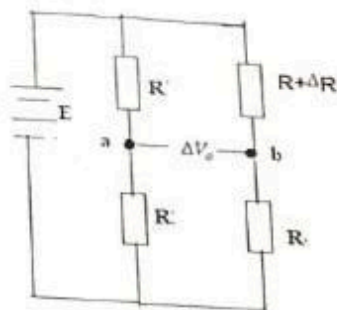


Figure Q4a

- b) For the circuit of Figure Q4b, derive the expression for the output voltage (6 marks)

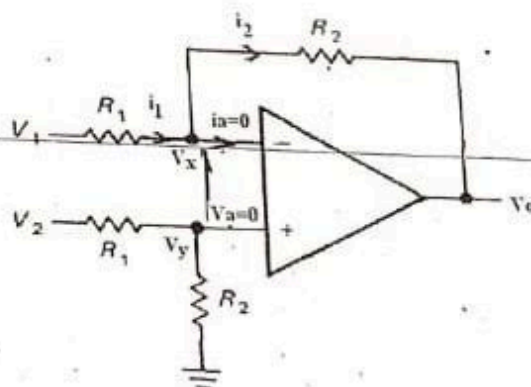


Figure Q4b

- c) Figure Q4c shows an instrumentation amplifier with a potentiometer $R_A = 200 \text{ k}\Omega$ when R_A is varied, Determine
- the minimum output voltage V_o
 - the maximum output voltage V_o
 - the output when R_A is set at the middle

(4marks)